

How to use LinearModelR: Linear Model Shiny App

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This App serves as a wrapper for the R package *limma*. enabling complex, multivariate statistical modeling in an accessible and user-friendly way. If you intend to use this App for data analysis which will later be published, please cite Brunelli *et al.*, 2026 (DOI HERE) AND Ritchie *et al.*, 2015 (10.1093/nar/gkv007).

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Before getting started

1. Download the latest version of R: <https://cran.r-project.org/>
2. Download the RStudio IDE: <https://docs.posit.co/ide/user/#rstudio-ide-oss-downloads>
3. Download the following packages (if you have not already):

```
if (!requireNamespace("BiocManager", quietly = TRUE)) {  
  install.packages("BiocManager")  
BiocManager::install(c("shiny", "limma", "cmapR"))
```

4. Load the packages:

```
Library(c("shiny", "limma", "cmapR"))
```

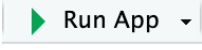
Downloading and setting up the App

1. There are two ways to access the App: manual download of the raw file or accessing the app through Github.
2. *Recommended* Access through Github:

- a. Open a new R session in RStudio. In the R console, run the following code block:

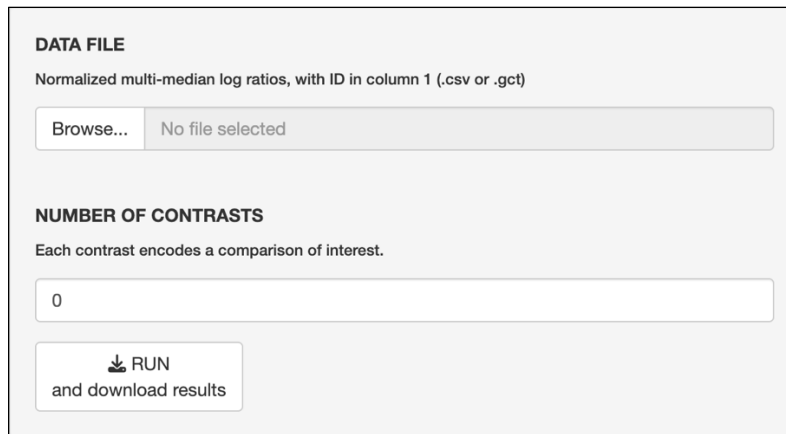
```
library(shiny)

shiny::runGitHub("LinearModelR", "melinabrunelli")
```

- b. The app interface should automatically pop up within a few seconds.
3. Manual download:
 - a. Navigate to <https://github.com/melinabrunelli/LinearModelR>.
 - b. Click on the app.R file link. Click on the “Download raw file” button in the top right corner. The App will immediately start downloading to your computer.
 - c. Store the App script in an easily accessible place on your local computer. It does not need to be in the same directory as your data files or R project files.
 - d. Open a new R session in RStudio, then open the LinearModelR.R file.
 - e. Click the  button in the top right corner of the script editor window. The app interface should pop up within a few seconds.

Uploading and configuring your data

1. First, you must upload your data file. On the left-hand side, you will see the following:



The screenshot shows the 'DATA FILE' section of the app interface. It includes a description: 'Normalized multi-median log ratios, with ID in column 1 (.csv or .gct)'. Below this is a file selection area with a 'Browse...' button and the text 'No file selected'. The next section is 'NUMBER OF CONTRASTS', with a description: 'Each contrast encodes a comparison of interest.' Below this is a text input field containing the number '0'. At the bottom, there is a button labeled 'RUN and download results' with a download icon.

2. Before uploading your data file, make sure that it is either a .csv or .gct file. This file should have an ID column (accession number, gene symbol) as the first column and the subsequent columns as normalized abundance values ($\log_2$ transformed intensities, $\log_2$ ratios, etc.). Example:

A	B	C	D	E	F	G	H	I
ID	LFQ.intensit y.FLAG.DMS O.FLAG.01	LFQ.intensit y.FLAG.DMS O.FLAG.02	LFQ.intensit y.FLAG.DMS O.FLAG.03	LFQ.intensit y.FLAG.OSM I.FLAG.01	LFQ.intensit y.FLAG.OSM I.FLAG.02	LFQ.intensit y.FLAG.OSM I.FLAG.03	LFQ.intensit y.STREP.DM SO.FLAG.01	LFQ.intensit y.STREP.DM SO.FLAG.02
Q4VAA2	24.823574	25.7513219	25.6340343	26.7359249	26.0698006	27.924991	25.7682403	25.213931
P83940	27.2779796	27.5575453	27.0454456	27.1540861	27.2768674	25.8897897	27.2769757	27.0848405
F8VQC7	25.9711241	25.8472337	26.013103	25.6736467	25.04385	25.1581143	25.8303841	25.3819585
E9PZ86	27.167208	26.9261625	27.3405445	26.7288849	26.9802033	25.1218805	24.7918122	24.6147402
Q3UQA7	25.7752157	25.7068145	26.1995002	26.9710929	26.699692	26.701703	25.6585931	25.5960997
P97789	26.7604737	26.4810705	26.534248	26.6185993	26.5531044	26.9717435	26.5943583	26.5102332
Q9CQC6	26.2577746	25.8428978	26.0350762	27.4907252	26.5144344	27.2985884	25.8843587	26.1536846
P11276	25.1135214	25.4046926	25.2142096	25.6046099	24.3595237	24.9693403	25.7119381	25.5078384
Q8BXX8	27.6136388	27.2971662	27.292872	27.2603574	26.7481465	27.805346	26.7007636	26.5925981

- Ensure that “Upload complete” has appeared after uploading your file before proceeding.
 - Note that you may have other columns that contain metadata or other samples, for example, but these cannot be the first column, and you will need to specify later that these are not sample columns.
 - An error will be thrown if you have spaces or special characters in your column headers.
 - If you upload a file that is not .csv or .gct, you will see error messages on the right-hand side: “Not a supported data file! csv and gct only.”
- Optional: set your number of contrasts. If you have more than two main comparisons (e.g., mutant vs. WT vs. control) or more than two secondary effects (e.g., treatment 1 vs. treatment 2 vs. control), you will have more than one contrast that can be made. If you only want the app to make a subset out of all possible contrasts, then indicate the appropriate number.
 - If you can only make one contrast, simply enter “1”.
 - [Samples] If you want all of the columns to be considered as samples, check the “Use all columns as samples.” box. If you have columns that do not represent samples, make sure to un-check this box, and then un-check those columns within the pop-up that appears.
 - Note that the App automatically considers the first column to be the ID column, so it will not show up here.
 - If you indicate any non-numeric columns as samples, an error will be thrown in the “Experimental Design” section and will not allow you to run the analysis.
 - [Main Comparisons] Your main comparisons will be the first contrast that is made. For AP-MS, your main comparison is enrichment (e.g., fOGT_{FLAG}) versus control (e.g., stOGT_{FLAG}). Indicate which columns in your data file correspond to which group by checking the corresponding boxes like so:

MAIN COMPARISONS

Enter main comparisons separated by commas, with reference group listed last. Eg: MUT, WT

fOGT, stOGT

fOGT ☒ LFQ.intensity.FLAG.DMSO.FLAG.01 ☒ LFQ.intensity.FLAG.DMSO.FLAG.02 ☒ LFQ.intensity.FLAG.DMSO.FLAG.03 ☒ LFQ.intensity.FLAG.OSMI.FLAG.01 ☒ LFQ.intensity.FLAG.OSMI.FLAG.02 ☒ LFQ.intensity.FLAG.OSMI.FLAG.03 ☐ LFQ.intensity.STREP.DMSO.FLAG.01 ☐ LFQ.intensity.STREP.DMSO.FLAG.02 ☐ LFQ.intensity.STREP.DMSO.FLAG.03 ☐ LFQ.intensity.STREP.OSMI.FLAG.01 ☐ LFQ.intensity.STREP.OSMI.FLAG.02 ☐ LFQ.intensity.STREP.OSMI.FLAG.03	stOGT ☐ LFQ.intensity.FLAG.DMSO.FLAG.01 ☐ LFQ.intensity.FLAG.DMSO.FLAG.02 ☐ LFQ.intensity.FLAG.DMSO.FLAG.03 ☐ LFQ.intensity.FLAG.OSMI.FLAG.01 ☐ LFQ.intensity.FLAG.OSMI.FLAG.02 ☐ LFQ.intensity.FLAG.OSMI.FLAG.03 ☒ LFQ.intensity.STREP.DMSO.FLAG.01 ☒ LFQ.intensity.STREP.DMSO.FLAG.02 ☒ LFQ.intensity.STREP.DMSO.FLAG.03 ☒ LFQ.intensity.STREP.OSMI.FLAG.01 ☒ LFQ.intensity.STREP.OSMI.FLAG.02 ☒ LFQ.intensity.STREP.OSMI.FLAG.03
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- a. If you have more than two groups, just ensure that the control is last.
6. [Secondary Effects] Secondary effects will be the second contrast that is made. Here, this is DMSO versus OSMI-4 treatment. The control treatment should be listed last. Format your groups in the same manner as was done in the Main Comparisons section.

SECONDARY EFFECTS (optional)

Enter secondary effects separated by commas, with reference treatment listed last. Eg: TRT, CTRL

OSMI4, DMSO

OSMI4 ☐ LFQ.intensity.FLAG.DMSO.FLAG.01 ☐ LFQ.intensity.FLAG.DMSO.FLAG.02 ☐ LFQ.intensity.FLAG.DMSO.FLAG.03 ☒ LFQ.intensity.FLAG.OSMI.FLAG.01 ☒ LFQ.intensity.FLAG.OSMI.FLAG.02 ☒ LFQ.intensity.FLAG.OSMI.FLAG.03 ☐ LFQ.intensity.STREP.DMSO.FLAG.01 ☐ LFQ.intensity.STREP.DMSO.FLAG.02 ☐ LFQ.intensity.STREP.DMSO.FLAG.03 ☒ LFQ.intensity.STREP.OSMI.FLAG.01 ☒ LFQ.intensity.STREP.OSMI.FLAG.02 ☒ LFQ.intensity.STREP.OSMI.FLAG.03	DMSO ☒ LFQ.intensity.FLAG.DMSO.FLAG.01 ☒ LFQ.intensity.FLAG.DMSO.FLAG.02 ☒ LFQ.intensity.FLAG.DMSO.FLAG.03 ☐ LFQ.intensity.FLAG.OSMI.FLAG.01 ☐ LFQ.intensity.FLAG.OSMI.FLAG.02 ☐ LFQ.intensity.FLAG.OSMI.FLAG.03 ☒ LFQ.intensity.STREP.DMSO.FLAG.01 ☒ LFQ.intensity.STREP.DMSO.FLAG.02 ☒ LFQ.intensity.STREP.DMSO.FLAG.03 ☐ LFQ.intensity.STREP.OSMI.FLAG.01 ☐ LFQ.intensity.STREP.OSMI.FLAG.02 ☐ LFQ.intensity.STREP.OSMI.FLAG.03
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- a. This step is optional: you may simply contrast your main comparisons, which is essentially a moderated T-test with BH multiple test correction.
7. [Contrasts] If your experimental design was set up correctly, you should see a contrast automatically populate.

CONTRASTS (optional)

Contrast 1

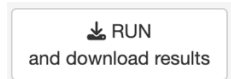
(fOGT.OSMI4-fOGT.DMSO)-(stOGT.OSMI4-stOGT.DMSO) ▼

- a. You should see your controls listed last (e.g. stOGT and DMSO are after fOGT and OSMI-4).

8. If your experimental design was set up correctly, a message will appear at the bottom, indicating your data can be analyzed:

EXPERIMENTAL DESIGN

Linear model experimental design is producible! Hit the 'RUN' button to download and view results.



9. Hit the button in the top left panel. This will prompt you to choose where your results file will be saved and rename the file if you wish.
 - a. If there are any issues that arise, you may see “downloadData.html” as the file name instead. This means the App will not process your results. Check the console in RStudio for errors.
10. Your results file should be available within a few minutes, depending on how many samples and contrasts you have.

Analyzing your results

1. Three new columns per contrast, in addition to the original columns in your data file, will be in the results file: [contrast].logFC, [contrast].P.Value, and [contrast].adj.P.Val
 - a. Note that the column header for the ID column may have been written over and is now blank.
2. The data may now be analyzed just like any other pairwise comparison! Specifically, you are now directly comparing the two groups in the first set of parentheses (e.g. fOGT.OSMI4 – fOGT.DMSO) and the other control groups have been taken into account with the model.
3. Questions? Please post on our Github page (<https://github.com/melinabrunelli/LinearModelR>).